Another incident that conceivably might represent a paradoxical ESS was recorded in a letter to *The Times* (of London, 7 December 1977) from a Mr James Dawson: 'For some years I have noticed that a gull using a flag pole as a vantage point invariably makes way for another gull wishing to alight on the post and this irrespectively of the size of the two birds.'

The most satisfying example of a paradoxical strategy known to me involves domestic pigs in a Skinner box. The strategy is stable in the same sense as an ESS, but it is better called a DSS ('developmentally stable strategy') because it arises during the animals' own lifetimes rather than over evolutionary time. A Skinner box is an apparatus in which an animal learns to feed itself by pressing a lever, food then being automatically delivered down a chute. Experimental psychologists are accustomed to putting pigeons or rats in small Skinner boxes, where they soon learn to press delicate little levers for a food reward. Pigs can learn the same thing in a scaled-up Skinner box with a very undelicate snout-lever (I saw a research film of this many years ago and I well recall almost dying of laughter). B. A. Baldwin and G. B. Meese trained pigs in a Skinner sty, but there is an added twist to the tale. The snout-lever was at one end of the sty; the food dispenser at the other. So the pig had to press the lever, then race up to the other end of the sty to get the food, then rush back to the lever, and so on. This sounds all very well, but Baldwin and Meese put *pairs* of pigs into the apparatus. It now became possible for one pig to exploit the other. The 'slave' pig rushed back and forth pressing the bar. The 'master' pig sat by the food chute and ate the food as it was dispensed. Pairs of pigs did indeed settle down into a stable 'master/slave' pattern of this kind, one working and running, the other doing most of the eating.

Now for the paradox. The labels 'master' and 'slave' turned out to be all topsy-turvy. Whenever a pair of pigs settled down to a stable pattern, the pig that ended up playing the 'master' or 'exploiting' role was the pig that, in all other ways, was subordinate. The so-called 'slave' pig, the one that did all the work, was the pig that was usually dominant. Anybody knowing the pigs would have predicted that, on the contrary, the dominant pig would have been the master, doing most of the eating; the subordinate pig should have been the hard-working and scarcely-eating slave.

How could this paradoxical reversal arise? It is easy to understand, once you start thinking in terms of stable strategies. All that we have to do is scale the idea down from evolutionary time to developmental time, the time-scale on which a relationship between two individuals develops. The strategy 'If dominant, sit by the food trough; if subordinate, work the lever' sounds sensible, but would not be stable. The subordinate pig, having pressed the lever, would come sprinting over, only to find the dominant pig with its front feet firmly in the trough and impossible to dislodge. The subordinate pig would soon give up pressing the lever, for the habit would never be rewarded. But now consider the reverse strategy: 'If dominant, work the lever; if subordinate, sit by the food trough.' This would be stable, even though it has the paradoxical result that the subordinate pig gets most of the food. All that is necessary is that there should be *some* food left for the dominant pig when he charges up from the other end of the sty. As soon as he arrives, he has no difficulty in tossing the subordinate pig out of the trough. As long as there is a crumb left to reward him, his habit of working the lever, and thereby inadvertently stuffing the subordinate pig, will persist. And the subordinate pig's habit of reclining idly by the trough is rewarded too. So the whole 'strategy', 'If dominant behave as a "slave", if subordinate behave as a "master" ', is rewarded and therefore stable.

p. 106 *… a kind of dominance hierarchy [in crickets] …*

Ted Burk, then my graduate student, found further evidence for this kind of pseudo-dominance hierarchy in crickets. He also showed that a male cricket is more likely to court females if he has recently won a fight against another male. This should be called the 'Duke of Marlborough Effect', after the following entry in the diary of the first Duchess of Marlborough: 'His Grace returned from the wars today and pleasured me twice in his top-boots.' An alternative name